SECP3843

SPECIAL TOPICS IN DATA ENGINEERING (WBL)

Semester 2 Session 2025/2026

End-To-End Data Engineering Project Using Microsoft Azure

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Section: 01

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1. Project Overview

This project simulates a real-life business problem faced by organizations where it is difficult to understand customer demographics, particularly gender distribution and how it influences product purchasing behavior. Understanding customer behavior is crucial to make effective business decisions. However, fragmented data systems make it difficult to get meaningful insights from raw operational data.

A Key Performance Indicator (KPI) dashboard is created using Power BI to allow organizations to monitor sales performance segmented by gender and product category. The dashboard displays the total products sold, total sales revenue, gender split between the customer and the ability to filter data by category, gender, and date. By using this dashboard, business teams are now able to identify which customer segments are driving revenue in specific product categories.

The data source for this project is the AdventureWorksLT database, which is a free sample retail database provided by Microsoft for on-premise Microsoft SQL Server. The goal of this project is to design an end-to-end data engineering pipeline using Microsoft Azure services such as Microsoft Data Factory, Databricks, Synapse and Power BI to extract raw data from the SQL Server, transform the data and create an interactive Power BI dashboard that helps stakeholders have access to up-to-date insights.

2. Description on the solution

To address the business problem, Medallion Architecture is used for creating the data pipeline. It is a layered data pipeline that organizes data into three stages which are Bronze, Silver and Gold.

In the Bronze layer, raw data is extracted directly from the on-premises SQL Server and stored as Parquet files in Azure Data Lake Storage Gen2 using Microsoft Data Factory. Then, transformations are applied to the raw data, specifically converting timestamp columns such ModifiedDate, Orderdate, DueDate, ShipDate into a clean date format and is stored in the Silver layer. In the Gold layer, further transformations are applied including renaming columns using Synapse. This final layer ensures that the column names are readable and the data is clean so it can be optimized for Power BI usage.

It also uses automation to run daily using Microsoft Data Factory. This means that stakeholders do not need to manually trigger to refresh the dashboard and the dashboard will always reflect the most current data. In terms of security, all credentials and connection secrets are stored in Azure Key Vault. Role Based Access Control (RBAC) is also applied across Azure services to ensure that only authorized users and services can access certain resources. All this design choice ensures that the data pipeline is secure and scalable.

3. Technology Use

The solution was built using Microsoft technologies that have specific roles within the pipeline. The Table 3.1 below summarizes the technologies used and their purpose in this project.

Table 3.1 Microsoft Azure Services and Supporting Tools Used in the Project

Technology	**Purpose**
SQL Server Express	On-premise database that hosts the AdventureWorksLT dataset
SQL Server Management Studio (SSMS)	Used to interact with and query the on-premise database
Azure Data Factory (ADF)	Orchestrates and automates the data pipeline
Azure Data Lake Storage Gen2 (ADLS)	Stores raw and transformed data across Bronze, Silver and Gold containers
Azure Databricks	Perform data transformations between layers using PySpark notebooks
Azure Synapse Analytics	Load Gold layer data into structured SQL views for querying
Azure Key Vault	Securely stores database credentials and connection secrets
Microsoft Entra ID	Manages access control and security groups across Azure services
Microsoft Power BI	Builds the KPI dashboard and visualizes analytical insights

4. Data Pipeline Architecture

Data pipeline architecture is the design of how data moves through a system from where it is generated to where it is stored, processed, and used. The architecture is designed to ensure efficient data ingestion, transformation, storage, and visualization while maintaining data quality and scalability. Figure 4.1 illustrates the architecture of the data pipeline used in this project, which follows a medallion architecture pattern and uses multiple Azure services.

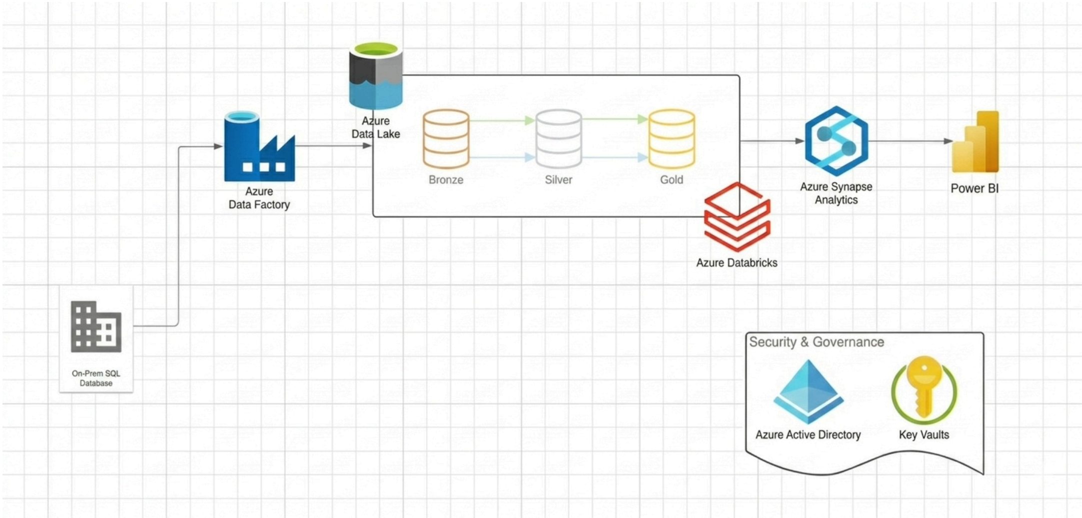

Figure 4.1 End-to-End Data Pipeline Architecture using Microsoft Azure

The pipeline begins with the AdventureWorksLT database hosted on an on-premises SQL Server, which serves as the primary data source containing sales and customer information. Azure Data Factory (ADF) acts as the central orchestration layer that manages the entire data flow from ingestion to transformation.

When the ADF pipeline shown in Figure 4.2 is triggered, data is extracted from the on-premises SQL Server and loaded into Azure Data Lake Storage Gen2, shown in Figure 4.3. This data is stored in the Bronze container as raw Parquet files. Then, it triggers Azure Databricks notebooks that perform the data transformation tasks. The data from the Bronze container is cleaned and processed by the Bronze to Silver notebook, which converts timestamp columns such as ModifiedDate, OrderDate, DueDate, and ShipDate into a proper date format, and the output is stored in the Silver container. After that, the Silver to Gold notebook runs to rename the columns into a more readable and consistent format so the data is ready for reporting, and the result is stored in the Gold container. These notebooks are integrated directly into the

ADF pipeline activities to allow the transformation process to be executed as part of the same workflow.

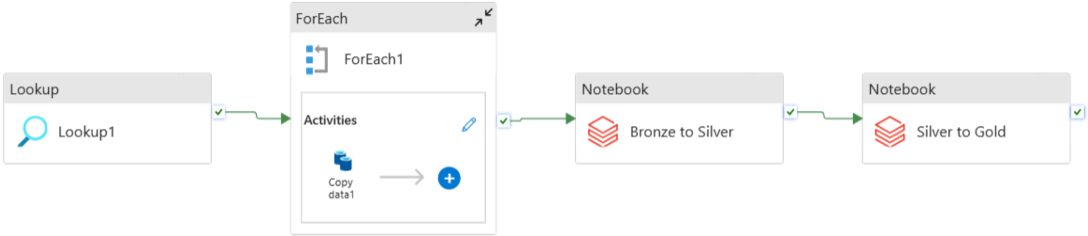

Figure 4.2 Azure Data Factory Pipeline

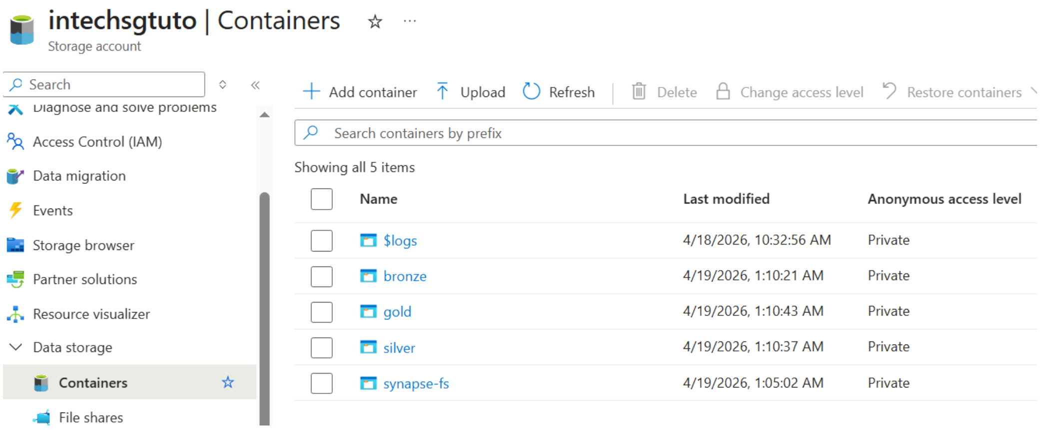

Figure 4.3 Azure Data Lake Storage Gen2 Containers

The final stage of the pipeline prepares the data for analytical use. The processed datasets are further refined and stored in the Gold layer before being integrated with Azure Synapse Analytics. Synapse provides a structured environment where the curated datasets can be queried efficiently using SQL. These datasets are then used as the source for building analytical dashboards in Microsoft Power BI.

In addition to data processing, security and governance is addressed using Azure Key Vault to securely store sensitive information such as database credentials and connection strings. Instead of hardcoding these secrets in Azure Data Factory or Databricks notebooks, Key Vault provides a secure central storage that reduces the risk of exposure.

5. MS Power BI dashboard

The Microsoft Power BI dashboard serves as the visualization layer of the data pipeline and provides stakeholders with an interactive platform for exploring sales performance and customer demographics. The dashboard is designed to present key business indicators in a clear and intuitive format so that users can quickly interpret the data. Figure 5.1 shows the dashboard results from the pipeline.

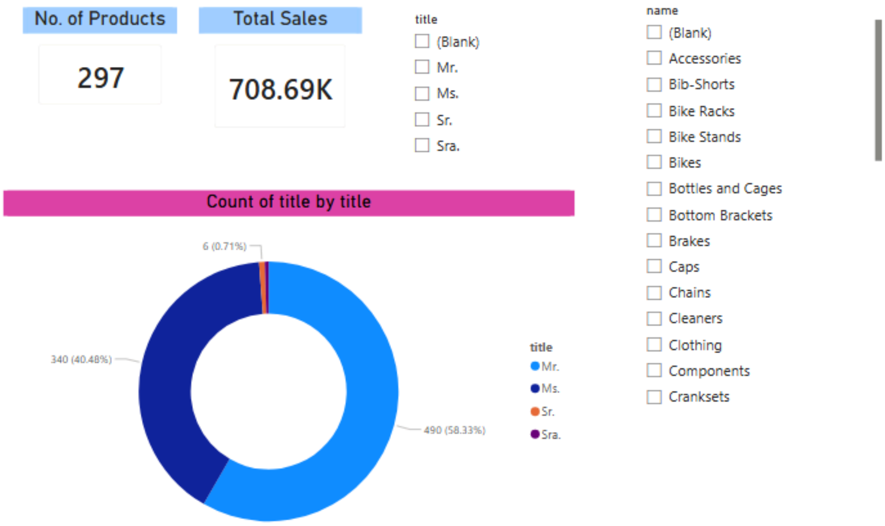

Figure 5.1 Sales Dashboard Results in Power BI

The dashboard in Figure 5.1 highlights several important metrics, including total sales revenue and total number of products sold, which provide a summary of the organization's sales performance. It also presents customer gender distribution to enable analysis of purchasing behavior across different demographic groups. It also displays a sales-by-product-category visualization that offers deeper insight into which categories generate the highest revenue and which customer segments contribute most to those sales. A key advantage of using Power BI is its interactive filtering capability. This dashboard allows users to filter the data by gender and product category and reflect it on the no. of products, total sales and gender count.

6. Reflection

This section shows the reflection of both team members after completing the project:

Najma Shakirah binti Shahrulzaman

After completing this project, I have learnt the core fundamentals of designing and implementing a data pipeline. This project was a step up for my career preparation as a data engineering student as designing pipelines is a main part of data engineering. I learned that security practices are also important when designing a data pipeline in cloud environments to avoid any.. The most significant challenge for me is configuring the data connection between all different Microsoft Azure services. It required understanding what different resources do and how to connect it with one another. Using documentation and available resources from YouTube, I was able to solve the connection issue. From a teamwork perspective, both of us have helped each other find a solution whenever an issue arises during this project. Overall, this project has strengthened our technical knowledge in cloud data engineering under constraints of real project delivery.

Nawwarah Auni binti Nazrudin

Through completing this end-to-end data engineering tutorial using Microsoft Azure, I have gained a stronger understanding of how a data pipeline is designed and implemented from data ingestion to visualization. I now have a clearer view of how Azure Data Factory, Databricks, Azure Data Lake Storage, and Power BI work together. Each tool has a specific role in moving and transforming data efficiently. One of the key things I learned is how to properly implement the Medallion Architecture, which separates data into Bronze, Silver, and Gold layers to ensure better quality and usability of data. This approach makes the pipeline more organised and scalable. I would like to apply this concept in my PPG project. From this experience, I feel more confident in handling data engineering workflows and it has also given me better insight into a potential career path in data engineering and cloud computing.